

```

In [1]: import numpy as np

def stab(n,type):
    a = np.zeros([n,3],dtype=type)
    a[0,0] = 1.0
    a[1,0] = (np.sqrt(5) - 1) / 2
    phi_1 = a[1,0]
    phi_im = a[0,0]
    phi_i = phi_1

    for i in range(2,n):
        phi_ip = phi_im - phi_i
        phi_im = phi_i
        phi_i = phi_ip
        a[i,:] = [phi_i, phi_1**i, 1-phi_i/phi_1**i]
        print('%3d %12.4e %12.4e %12.4e ' % (i, phi_1**i, phi_i, 1-phi_i/phi_1
**i))

    return a

```

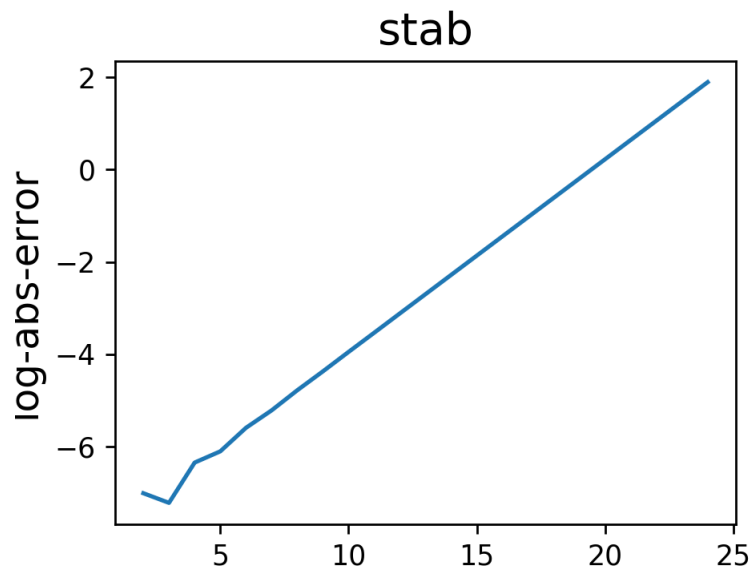
```

In [2]: golden = stab(25, 'float32')

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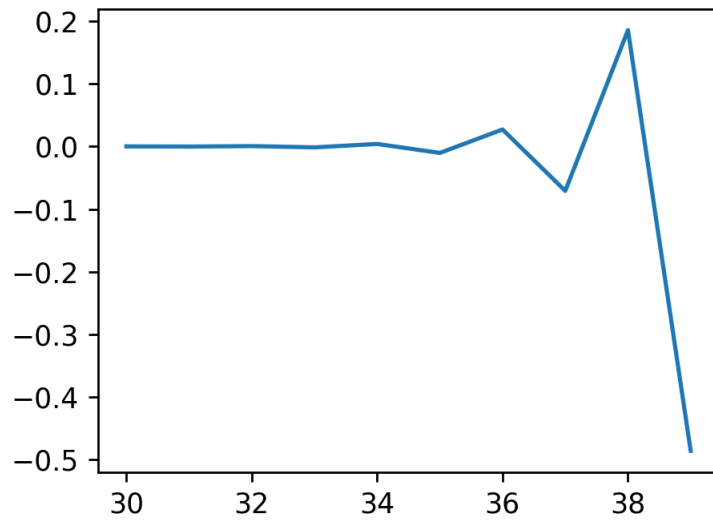
2	3.8197e-01	3.8197e-01	9.6096e-08
3	2.3607e-01	2.3607e-01	-5.9391e-08
4	1.4590e-01	1.4590e-01	4.4378e-07
5	9.0170e-02	9.0170e-02	-7.7744e-07
6	5.5728e-02	5.5728e-02	2.5158e-06
7	3.4442e-02	3.4442e-02	-6.0100e-06
8	2.1286e-02	2.1286e-02	1.6407e-05
9	1.3156e-02	1.3156e-02	-4.2185e-05
10	8.1306e-03	8.1297e-03	1.1131e-04
11	5.0250e-03	5.0265e-03	-2.9044e-04
12	3.1056e-03	3.1033e-03	7.6145e-04
13	1.9194e-03	1.9232e-03	-1.9924e-03
14	1.1862e-03	1.1801e-03	5.2173e-03
15	7.3314e-04	7.4315e-04	-1.3658e-02
16	4.5310e-04	4.3690e-04	3.5758e-02
17	2.8003e-04	3.0625e-04	-9.3614e-02
18	1.7307e-04	1.3065e-04	2.4509e-01
19	1.0696e-04	1.7560e-04	-6.4164e-01
20	6.6107e-05	-4.4942e-05	1.6798e+00
21	4.0856e-05	2.2054e-04	-4.3979e+00
22	2.5251e-05	-2.6548e-04	1.1514e+01
23	1.5606e-05	4.8602e-04	-3.0143e+01
24	9.6449e-06	-7.5150e-04	7.8916e+01

```
In [3]: %matplotlib notebook
from matplotlib import pyplot as plt
import numpy as np
plt.figure(figsize=(4,3))
plt.plot(range(2,25),np.log10(abs(golden[2:25,2])))
plt.title('stab',fontsize=16)
plt.xlabel('n',fontsize=12)
plt.ylabel('log-abs-error',fontsize=14)
plt.show()
```



```
In [28]: %matplotlib notebook
from matplotlib import pyplot as plt
import numpy as np
plt.figure(figsize=(4,3))
plt.plot(range(30,40),stab(41,'float64')[30:40,2])
plt.title('stab',fontsize=16)
plt.xlabel('n',fontsize=14)
plt.ylabel('error',fontsize=14)
plt.show()
```

stab



2	3.8197e-01	3.8197e-01	3.3307e-16
3	2.3607e-01	2.3607e-01	-2.2204e-16
4	1.4590e-01	1.4590e-01	1.5543e-15
5	9.0170e-02	9.0170e-02	-2.6645e-15
6	5.5728e-02	5.5728e-02	8.3267e-15
7	3.4442e-02	3.4442e-02	-1.9984e-14
8	2.1286e-02	2.1286e-02	5.4290e-14
9	1.3156e-02	1.3156e-02	-1.3967e-13
10	8.1306e-03	8.1306e-03	3.6826e-13
11	5.0250e-03	5.0250e-03	-9.6101e-13
12	3.1056e-03	3.1056e-03	2.5198e-12
13	1.9194e-03	1.9194e-03	-6.5932e-12
14	1.1862e-03	1.1862e-03	1.7265e-11
15	7.3314e-04	7.3314e-04	-4.5196e-11
16	4.5310e-04	4.5310e-04	1.1833e-10
17	2.8003e-04	2.8003e-04	-3.0979e-10
18	1.7307e-04	1.7307e-04	8.1104e-10
19	1.0696e-04	1.0696e-04	-2.1233e-09
20	6.6107e-05	6.6107e-05	5.5589e-09
21	4.0856e-05	4.0856e-05	-1.4553e-08
22	2.5251e-05	2.5251e-05	3.8101e-08
23	1.5606e-05	1.5606e-05	-9.9751e-08
24	9.6449e-06	9.6449e-06	2.6115e-07
25	5.9609e-06	5.9609e-06	-6.8370e-07
26	3.6840e-06	3.6840e-06	1.7900e-06
27	2.2768e-06	2.2769e-06	-4.6862e-06
28	1.4072e-06	1.4072e-06	1.2269e-05
29	8.6968e-07	8.6971e-07	-3.2119e-05
30	5.3749e-07	5.3745e-07	8.4090e-05
31	3.3219e-07	3.3226e-07	-2.2015e-04
32	2.0530e-07	2.0518e-07	5.7636e-04
33	1.2688e-07	1.2708e-07	-1.5089e-03
34	7.8419e-08	7.8109e-08	3.9504e-03
35	4.8465e-08	4.8967e-08	-1.0342e-02
36	2.9953e-08	2.9142e-08	2.7077e-02
37	1.8512e-08	1.9824e-08	-7.0887e-02
38	1.1441e-08	9.3178e-09	1.8559e-01
39	7.0710e-09	1.0507e-08	-4.8587e-01
40	4.3701e-09	-1.1888e-09	1.2720e+00

In []: